Quantum Approaches to Joint-Loss Modelling for Natural Catastrophe Risk

Challenge Description

1. Context and Relevance

Catastrophe risk products are financial instruments that move the cost of disaster losses away from governments and onto capital markets or insurers before a disaster happens. They include catastrophe bonds, parametric insurance and contingent credit lines. In a catastrophe bond, investors receive a regular coupon in exchange for agreeing to lose some or all of their principal if a defined disaster occurs. Parametric products work differently from conventional insurance. They pay out based on an objective, pre-agreed trigger, such as earthquake magnitude, ground shaking intensity or rainfall levels, instead of a lengthy assessment of actual damage. This lets funds reach affected governments within days or weeks, when liquidity matters most for emergency response and recovery. For developing economies with limited fiscal buffers, these instruments can reduce reliance on post-disaster borrowing and budget reallocation. However, their pricing and reliability depend on catastrophe models that estimate how often extreme events occur, how severe they are and where they hit.

ADB is expanding its portfolio in the space of such catastrophe risk products, including a recent US$160 million Disaster Relief Bond for the Kyrgyz Republic and Tajikistan. The catastrophe models behind these products face two persistent problems: only a couple of decades of high-quality disaster data exist, and the dependence between perils across countries is poorly constrained by history.

In the example to be studied in this challenge, the two countries share fault lines and glacier systems, so a single seismic episode could plausibly drive losses in both at once across various disaster modalities. Importantly, the joint tail – how extreme events co-move across borders – is what determines whether pooling risk across countries reduces cost, and whether parametric triggers pay out when real losses occur. However, the joint tail cannot be estimated from history alone because of limited records.

Relevance for Quantum. It is thus of interest how a bank like ADB can model the dependence between rare, extreme events across a multi-country portfolio where data is limited. Today, these calculations are done by exact enumeration methods if the model is small, or by Monte Carlo type simulations if it is not. Sampling error on these rare joint events is a well-known bottleneck. Quantum and quantum-inspired methods – such as quantum annealing, Quantum Boltzmann Machines, quantum-inspired sampling and tensor networks – may offer pathways to represent joint distributions more faithfully, or with fewer parameters, than classical models of comparable size. Whether this holds against strong classical baselines, at realistic resource counts, is the open question this challenge is designed to answer with evidence.

2. Challenge Overview

In this quantum challenge, teams are tasked with evaluating whether quantum/quantum-inspired methods can estimate the joint loss and payout risk of a multi-country disaster risk finance portfolio, and if yes, pushing the limits of how large a portfolio can be simulated while

maintaining accuracy benchmarks.

An open benchmark repository is provided with two worked examples, exact reference values and classical baselines to measure against. However, teams are not restricted to the shared approaches and are encouraged to be creative in their scientific methods.

Higher accuracy per query than the classical baselines on the benchmark today, and a credible scaling argument for larger portfolios, will be evaluated for successful challenge submissions.

Phase 1 Open registration. Teams propose their scientific approach to tackle the defined problem statement. Teams are expected to submit thorough ideation, literature evidence (if any) and a lightweight prototype to report estimates and resources required for full development by the end of Phase 1. A clear plan for Phase 2 activities should be outlined along with the submission. See the evaluation criteria for more details.

Phase 2 Selected teams from Phase 1 are accepted for Phase 2 and onboarded to the QAI Ventures sandbox. Teams will build a working, tested and reproducible prototype. Teams will be provided with exclusive GPU credits to show scalability of the approach. Teams submit a final report, codebase documentation and demo as final Phase 2 deliverables.

Teams are advised to refer to the website for the timelines and submission deadlines for each of the above Challenge Phases.

3. Detailed Problem Description

a. Mathematical Problem / Model Description

A portfolio covers several countries, each with a contract paying on one or more perils. A peril zone unit $u = 1, \ldots, U$ is the level at which loss or a trigger outcome is recorded between a country and peril pair. Let $X = (X_1, \ldots, X_U)$ be the joint outcome across units in a period, with distribution $p(x)$. Each country's contract maps outcomes to a payout $c_i(x)$, and the portfolio loss is $L = \sum_i c_i(x)$. If p is the dependence structure and f is the contract terms, every target can be represented as:

$$\mu \;=\; \mathbb{E}_p\big[f\big(c_1(X), \ldots, c_n(X)\big)\big] \;=\; \sum_x p(x)\, f\big(c_1(x), \ldots, c_n(x)\big). \tag{1}$$

b. Demo Example & Benchmarks

Teams may use the following example as a reference point to understand the classical approach and get started. Teams should provide a comparison with the benchmark results in the examples below.

Consider two countries, the Kyrgyz Republic and Tajikistan. Each bond has USD 80m principal over a three-year term, split into a USD 70m earthquake sub-limit and a USD 10m precipitation sub-limit. Earthquake payouts come in quarter steps of the sub-limit (0%, 25%, 50%, 75%, 100%) and accumulate over the term up to the sub-limit. Precipitation pays its full sub-limit at most once per term.

1. For each country, the annual probability that the year's aggregate earthquake payout reaches at least 25%, 50%, 75% and 100% of the sub-limit is given in the dataset as below. Also given is the probability that the same earthquake triggers the other country's bond, given that it triggers one: Tajikistan given Kyrgyz Republic is 7.5%; Kyrgyz Republic given Tajikistan is 7.4%.

Table 1: Annual earthquake payout exceedance probabilities.

Country	≥25% (attachment)	≥50%	≥75%	100% (exhaustion)	Published annual EL
Kyrgyz Republic	3.883%	2.331%	1.790%	1.184%	2.296%
Tajikistan	3.925%	1.743%	1.332%	1.043%	2.011%

2. Model each country's annual earthquake payout as a compound Poisson process. Events arrive at total rate Λ and each pays k quarter steps ($k = 1, 2, 3, 4$) with rate λ_k. From the four tail probabilities $t_1, \ldots, t_4$, form the point probabilities of paying 0, 1, 2 or 3 quarters in a year: $p_0 = 1 - t_1$, $p_1 = t_1 - t_2$, $p_2 = t_2 - t_3$, $p_3 = t_3 - t_4$. Matching these to the compound Poisson probabilities gives:

$$\Lambda = -\ln p_0, \qquad \lambda_1 = \frac{p_1}{p_0}, \qquad \lambda_2 = \frac{p_2}{p_0} - \frac{\lambda_1^2}{2}, \tag{2}$$

$$\lambda_3 = \frac{p_3}{p_0} - \lambda_1\lambda_2 - \frac{\lambda_1^3}{6}, \qquad \lambda_4 = \Lambda - \lambda_1 - \lambda_2 - \lambda_3. \tag{3}$$

Resulting rates per year:

Table 2: Implied compound Poisson rates (per year).

Country	λ_1	λ_2	λ_3	λ_4	Λ
Kyrgyz Republic	0.016147	0.005498	0.006215	0.011743	0.039604
Tajikistan	0.022711	0.004020	0.002915	0.010395	0.040041

3. Some events may hit both countries. Set the shared-event rate c as the average of the two rates implied by the conditional probabilities, scaled by a scenario multiplier m (1 in the base case). Each country keeps its own total rate. Events hitting country 1 only arrive at rate $\Lambda_1 - c$, country 2 only at $\Lambda_2 - c$, and both at c. Each event's size in each country is drawn from $w_{i,k} = \lambda_{i,k}/\Lambda_i$, independently across the two countries even for shared events.

$$c = m \cdot \tfrac{1}{2}\,(0.075\,\Lambda_1 + 0.074\,\Lambda_2) = 0.0029667 \text{ per year (base case)} \tag{4}$$

4. Track the year's cumulative quarter steps (a, b) for the two countries, each capped at 4, as a continuous-time Markov chain on 25 states. From state (a, b), the transitions are:

   - to $(\min(a + k, 4),\, b)$ at rate $(\Lambda_1 - c)\, w_{1,k}$;
   - to $(a,\, \min(b + j, 4))$ at rate $(\Lambda_2 - c)\, w_{2,j}$;
   - to $(\min(a + k, 4),\, \min(b + j, 4))$ at rate $c\, w_{1,k}\, w_{2,j}$.

   With G the 25×25 generator matrix, the annual joint distribution is the first row of the matrix exponential $\exp(G)$, starting from $(0, 0)$. In the base case this gives a 92.62% chance of no earthquake payout in either country in a year.

5. Each country triggers precipitation with annual probability $q = 2\%$, so its Poisson rate is $\mu = -\ln(1 - q) = 0.020203$, times a scenario multiplier (1 in the base case). A shared rate c_r equal to 25% of the smaller country rate, here 0.005051, drives common triggers. The four outcomes are:

$$P_{00} = e^{-(\mu_1 + \mu_2 - c_r)}, \qquad P_{10} = e^{-\mu_2} - P_{00}, \tag{5}$$

$$P_{01} = e^{-\mu_1} - P_{00}, \qquad P_{11} = 1 - P_{00} - P_{10} - P_{01}. \tag{6}$$

In the base case: neither 96.526%, one country only 1.474% each, both 0.526%.

6. An annual state is (a, b, r_1, r_2): earthquake quarter steps for each country (0 to 4) and precipitation triggers (0 or 1), so $5 \times 5 \times 2 \times 2 = 100$ states. Earthquake and precipitation are independent, so each state's probability is the product of its earthquake probability (step 4) and its precipitation probability (step 5). The chance of no payout of any kind in a year is 89.40%.

7. Years are independent. Starting from $(0, 0, 0, 0)$, add each year's annual state and cap the running total at $(4, 4, 1, 1)$. Repeat for three years to get the exact distribution of capped term states. The exact method does this by direct convolution: three passes over 100×100 state pairs. Each term state maps to principal loss:

$$L_1 = 70\text{m} \cdot \frac{a}{4} + 10\text{m} \cdot r_1, \qquad L_2 = 70\text{m} \cdot \frac{b}{4} + 10\text{m} \cdot r_2, \qquad L = L_1 + L_2. \tag{7}$$

8. With $T =$ USD 80m, take expectations over the term distributions (five focus targets):

Table 3: Focus targets.

Target	Definition
`term_expected_principal_loss_usd`	$\mathbb{E}[L]$
`term_joint_attachment_probability`	$\Pr(L_1 > 0 \text{ and } L_2 > 0)$
`term_portfolio_loss_exceedance_probability`	$\Pr(L > T)$
`term_portfolio_stop_loss_usd`	$\mathbb{E}[\max(L - T, 0)]$
`term_both_bonds_exhausted_probability`	$\Pr(L_1 = 80\text{m} \text{ and } L_2 = 80\text{m})$

9. Estimate the targets by sampling. The sampling baselines treat a term as a path of three annual states:

   - **Monte Carlo.** Draw three uniforms per path and map each to an annual state by inverting the cumulative annual distribution.
   - **Scrambled Sobol.** Do the same with three-dimensional Sobol points: eight independent scrambles, averaged, with a t-interval on 7 degrees of freedom.
   - **Stratified sampling.** Split paths by the first year with any payout (year 1, 2 or 3; a path with no payout contributes exactly zero). Spread the budget across the three strata and weight each by $p_0^{k-1}(1 - p_0)$, where $p_0 = 89.40\%$.
   - **Importance sampling.** Draw each year from a proposal that is half the true annual distribution and half the distribution tilted by each state's principal loss. Reweight each path by the product of true-to-proposal probability ratios.

   The scenarios, each of which changes one input and keeps the rest of the base case, are:

Table 4: Benchmark scenarios.

Scenario	Change from base case
`adb-base`	None
`adb-independent`	No shared events: $m = 0$ and precipitation shared share $= 0$
`adb-common-shock`	Shared earthquake rate doubled: $m = 2$
`adb-rain-stress`	Precipitation rates doubled
`adb-rain-cover`	Sub-limits changed to USD 60m earthquake and USD 20m precipitation per country

10. Results:

Table 5: Exact reference values per scenario (three-year term).

Scenario	Expected principal loss	Both countries pay	Loss > USD 80m	Expected loss above USD 80m	Both bonds exhausted
`adb-base`	USD 10.06m	4.414%	1.205%	USD 282,355	0.0032%
`adb-independent`	USD 10.06m	2.716%	0.719%	USD 171,266	0.0004%
`adb-common-shock`	USD 10.06m	5.053%	1.607%	USD 385,043	0.0044%
`adb-rain-stress`	USD 11.17m	7.046%	1.322%	USD 306,286	0.0069%
`adb-rain-cover`	USD 9.97m	4.414%	0.700%	USD 170,927	0.0032%

Note how dependence leaves expected loss unchanged but moves every joint target. Removing shared events cuts the chance of both countries paying from 4.4% to 2.7%, and the chance of both bonds being exhausted by a factor of eight. Also, the exhaustion target is about 3 in 100,000 per term, and this is where the sampling methods struggle.

Exact enumeration takes about 0.2 milliseconds for the ADB model and hence runtime is not an issue. The bottleneck is accuracy in the rare joint tail at a fixed budget. It also grows with portfolio size. With two perils on the ADB grid, the annual state space is 10^n for n countries: 100 states for two, 100,000 for five. Exact enumeration then becomes impractical, and sampling error on joint-tail targets grows.

c. Proposed Quantum Approaches

Teams may use the following quantum approaches as a reference point to get started, but it is not mandatory. Teams may decide on other creative approaches.

- **Quantum Amplitude Amplification.** Load $p(x)$ into amplitudes, compute f reversibly into an objective qubit, and estimate the resulting amplitude. In the ideal case, error falls as $1/M$ in oracle calls, against $1/\sqrt{M}$ for Monte Carlo.
- **Quantum Sampling Methods.** Learn $p(x)$ with a quantum circuit Born machine, quantum GAN, quantum Boltzmann machine (gate-based or annealer-based) or quantum copula model. Use it to sample joint scenarios or as the loading step for quantum amplitude amplification.
- **Quantum Machine Learning.** Use quantum kernels as surrogates for payout functions, or to learn better importance-sampling proposals.
- **Other** quantum/quantum-inspired methods.

d. Available Hardware

Teams are expected to use their own resources for the Phase 1 submission. Teams selected for Phase 2 will receive exclusive access to GPU credits through QAI Ventures. A tutorial for GPU usage will be provided for selected teams in Phase 2.

4. Expected Deliverables

Phase 1

- **Scientific Proposal:** A well-written scientific document clearly articulating your proposed quantum methodology for tackling the problem statement. The method (quantum algorithm, any pre/post-processing, encoding techniques, QEC, etc.) should be explained in detail with proper citations wherever necessary.

- **Resource Estimation:** Breakdown of all compute resources (quantum & classical), including number of qubits, circuit depth, two-qubit gates, shots, error mitigation, and mapping to the ideal hardware modality.
- **Lightweight Prototype:** Simulation results from a minimum viable solution which works as a validation of your proposed solution. Include the entire source code, environment specifications, hardware/simulator details, and any AI use in the submission package.
- **Phase 2 Plan of Action:** Outline your team's plan to build if selected for Phase 2.
- **Team:** Who you are, and why your team is best suited to successfully deliver the solution to this challenge.

Phase 2

- **Working Prototype:** A functional model that takes data as input and produces a solution to the posed challenge. We recommend a step-by-step approach, beginning with a proof-of-concept for a very simple toy model before expanding to a full model, e.g. tackling larger data sets or problem instances.
- **Technical Documentation:** A detailed report explaining the quantum algorithms used, the data model, and the rationale for choosing the specific approach. The focus should be on the clear and deliberate use of methodology and the benchmarking process, rather than solely on high-performing results. Participants should benchmark their quantum model against standard classical solutions. If more than one model was developed during the challenge, the team may also benchmark against those.
- **Classical Bottleneck Analysis and Justification for Quantum Computing Approach:** An analysis of standard approaches and their bottlenecks. This is the basis for arguing why a quantum computing approach may be beneficial for the problem at hand.
- **Business Case:** A concise plan outlining commercial viability, target users, and market strategy. Think about how your solution could be commercialized, how you would approach commercialization, what the product would be, and who the customers would be. Also: in what other fields or problems could your solution be useful?

5. Resources

Suggested literature for additional reading

Context

- https://cabar.asia/en/earthquakes-in-central-asia-casualties-and-half-a-billion-damage-per-year-kyrgyzstan-loses-the-most
- https://www.adb.org/news/adb-issues-inaugural-disaster-relief-bonds-targeting-earthquake-and-extreme-precipitation-risk

Extreme events and dependence

- Embrechts, P., Klüppelberg, C., and Mikosch, T. (1997). *Modelling Extremal Events for Insurance and Finance*. Springer. Foundations of heavy-tailed loss modelling.
- Joe, H. (2014). *Dependence Modeling with Copulas*. CRC Press. Copulas, vines and tail dependence.

Classical sampling

- Glasserman, P. (2003). *Monte Carlo Methods in Financial Engineering*. Springer. Stratified and importance sampling, quasi-Monte Carlo.
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Quantum amplitude estimation and quantum finance

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- Montanaro, A. (2015). Quantum speedup of Monte Carlo methods. *Proceedings of the Royal Society A*, 471.
- Woerner, S., and Egger, D. J. (2019). Quantum risk analysis. *npj Quantum Information*, 5, 15.
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- Suzuki, Y., et al. (2020). Amplitude estimation without phase estimation. *Quantum Information Processing*, 19, 75.
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- Chakrabarti, S., et al. (2021). A threshold for quantum advantage in derivative pricing. *Quantum*, 5, 463. A model for honest end-to-end resource estimation.
- Herman, D., et al. (2023). Quantum computing for finance. *Nature Reviews Physics*, 5, 450. Survey of the field and its open problems.

Distribution loading and quantum generative models

- Grover, L., and Rudolph, T. (2002). Creating superpositions that correspond to efficiently integrable probability distributions. arXiv:quant-ph/0208112.
- Zoufal, C., Lucchi, A., and Woerner, S. (2019). Quantum generative adversarial networks for learning and loading random distributions. *npj Quantum Information*, 5, 103.
- Benedetti, M., et al. (2019). A generative modeling approach for benchmarking and training shallow quantum circuits. *npj Quantum Information*, 5, 45. Quantum circuit Born machines.
- Amin, M. H., et al. (2018). Quantum Boltzmann machine. *Physical Review X*, 8, 021050.
- Zhu, E. Y., et al. (2022). Generative quantum learning of joint probability distribution functions. *Physical Review Research*, 4, 043092. Copula-based quantum generative models, directly relevant to joint tails.

Tensor networks

- Han, Z.-Y., Wang, J., Fan, H., Wang, L., and Zhang, P. (2018). Unsupervised generative modeling using matrix product states. *Physical Review X*, 8, 031012.
- Orús, R. (2014). A practical introduction to tensor networks. *Annals of Physics*, 349, 117.